

A Technical Introduction to Multi-armed bandit

Section 1

$$\rho_{\pi}(T) = \sum_{t=1}^T \mu_{t^*} - E \pi_{\mu}[\sum_{t=1}^T r_t] = D(T) - E \pi_{\mu}[\sum_{t=1}^T r_t].$$

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Section 2

For each $t = 1, 2, \dots, T$. For each arm generate a random noise from an exponential distribution $\forall i: Z_i(t) \sim \text{Exp}(\eta)$

Section 3

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Section 4

Semi-uniform strategies Semi-uniform strategies were the earliest (and simplest) strategies discovered to approximately solve the bandit problem. All those strategies have in common a greedy behavior where the best lever (based on previous observations) is always pulled except when a (uniformly) random action is taken.

Section 5

Exp3 Source: EXP3 is a popular algorithm for adversarial multiarmed bandits, suggested and analyzed in this setting by Auer et al. [2002b]. Recently there was an increased interest in the performance of this algorithm in the stochastic setting, due to its new applications to stochastic multi-armed bandits with side information [Seldin et al., 2011] and to multi-armed bandits in the mixed stochastic-adversarial setting [Bubeck and Slivkins, 2012]. The paper presented an empirical evaluation and improved analysis of the performance of the EXP3 algorithm in the stochastic setting, as well as a modification of the EXP3 algorithm capable of achieving "logarithmic" regret in stochastic environment.

Section 6

4. For $j = 1, \dots, K$ set: $x_j(t) = \{x_j(t) / p_j(t) \text{ if } j = i(t), \text{ otherwise}\}$

Section 7

Add noise to each arm and pull the one with the highest value 3. Update value: $R_{I(t)}(t+1) = R_{I(t)}(t) + x_{I(t)}(t)$

Section 8

Bandit strategies A major breakthrough was the construction of optimal population selection strategies, or policies (that possess uniformly maximum convergence rate to the population with highest mean) in the work described below.

Section 9

$\omega_j(t+1) = \omega_j(t) \exp(\gamma x_j(t) / K)$

Section 10

Combinatorial bandit The Combinatorial Multiarmed Bandit (CMAB) problem arises when instead of a single discrete variable to choose from, an agent needs to choose values for a set of variables. Assuming each variable is discrete, the number of possible choices per iteration is exponential in the number of variables. Several CMAB settings have

been studied in the literature, from settings where the variables are binary to more general setting where each variable can take an arbitrary set of values.

Section 11

Variations A common formulation is the Binary multi-armed bandit or Bernoulli multi-armed bandit, which issues a reward of one with probability p , and otherwise a reward of zero. Another formulation of the multi-armed bandit has each arm representing an independent Markov machine. Each time a particular arm is played, the state of that machine advances to a new one, chosen according to the Markov state evolution probabilities. There is a reward depending on the current state of the machine. In a generalization called the "restless bandit problem", the states of non-played arms can also evolve over time. There has also been discussion of systems where the number of choices (about which arm to play) increases over time. Computer science researchers have studied multi-armed bandits under worst-case assumptions, obtaining algorithms to minimize regret in both finite and infinite (asymptotic) time horizons for both stochastic and non-stochastic arm payoffs.

Section 12

UCB-ALP algorithm: The framework of UCB-ALP is shown in the right figure. UCB-ALP is a simple algorithm that combines the UCB method with an Adaptive Linear Programming (ALP) algorithm, and can be easily deployed in practical systems. It is the first work that show how to achieve logarithmic regret in constrained contextual bandits. Although is devoted to a special case with single budget constraint and fixed cost, the results shed light on the design and analysis of algorithms for more general CCB problems.

Section 13

Online non-linear bandits **UCBogram algorithm:** The nonlinear reward functions are estimated using a piecewise constant estimator called a regressogram in nonparametric regression. Then, UCB is employed on each constant piece. Successive refinements of the partition of the context space are scheduled or chosen adaptively. **Generalized linear algorithms:** The reward distribution follows a generalized linear model, an extension to linear bandits. **KernelUCB algorithm:** a kernelized non-linear version of LinUCB, with efficient implementation and finite-time analysis. **Bandit Forest algorithm:** a random forest is built and analyzed w.r.t the random forest built knowing the joint distribution of contexts and rewards. **Oracle-based algorithm:** The algorithm reduces the contextual bandit problem into a series of supervised learning problem, and does not rely on typical realizability assumption on the reward function.